SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	Slot A / 2024	**Date:**	28-10-2026

Code of Conduct

I G. Poojitha (192425337) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Poojitha

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Scenario-Based

Qns

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
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Q1 – IMAGE ENHANCEMENT USING GREY-LEVEL TRANSFORMATIONS

1.1 Problem identification

The hospital X-ray system produces images that are too dark, making bones, soft tissues, and other anatomical structures difficult to distinguish. The immediate problem is that useful pixel intensities occupy a low or narrow part of the available grayscale range. A suitable point-processing method should increase brightness and expand the intensity differences in the range containing useful anatomical information.

1.2 Selected method: contrast stretching with brightness adjustment

Contrast stretching maps a limited input intensity range to a wider output range. A simple linear transformation is:

$$s = \text{clip}[a(r - r_min) + b, 0, 255]$$

Here, r is the original pixel intensity, s is the transformed intensity, a is the contrast gain, b is a brightness offset, and clip limits the output to the valid range 0–255. For this worked example, use a = 1.35, r_min = 20, and b = 20. Therefore, s = clip[1.35(r − 20) + 20, 0, 255]. The values are illustrative; a real X-ray should use parameters selected from its intensity distribution and clinical requirements.

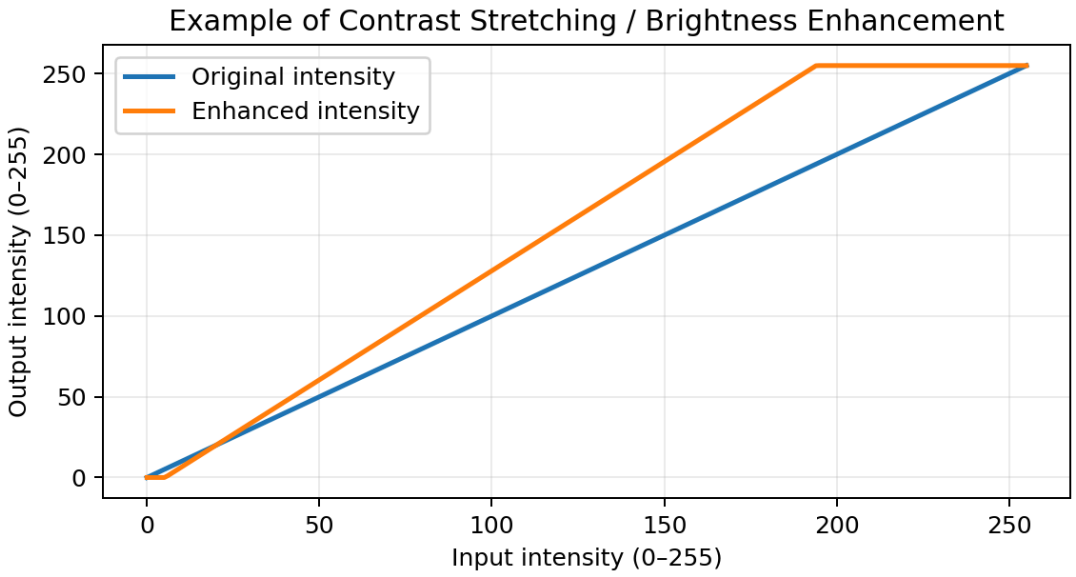


Figure 1. The enhanced curve maps much of the useful input range to a wider output range.

1.3 Worked pixel calculations

Assume that five selected pixels in a dark X-ray have original intensity values 20, 40, 60, 80, and 100. Applying the chosen transformation gives the following values. Rounding is to the nearest integer.

Pixel	Original r	Calculation	Transformed s
P1	20	$1.35(20 - 20) + 20$	20
P2	40	$1.35(40 - 20) + 20 = 47$	47
P3	60	$1.35(60 - 20) + 20 = 74$	74
P4	80	$1.35(80 - 20) + 20 = 101$	101
P5	100	$1.35(100 - 20) + 20 = 128$	128

Example for P4: the original intensity is r = 80. Subtracting 20 gives 60; multiplying by 1.35 gives 81; adding 20 gives 101. Hence, the pixel changes from intensity 80 to 101. The output remains within the allowed 8-bit range.

1.4 Interpretation of the calculation

The transformation brightens pixels above the selected lower reference level and increases the separation between nearby intensity values. For example, the input difference between pixels with intensities 40 and 60 is 20. Their transformed values are 47 and 74, so the output difference is 27. The contrast gain of 1.35 therefore makes that difference more visible. The lower reference pixel at intensity 20 remains at 20 in this example.

1.5 How it improves X-ray visibility

- Anatomical structures may become easier to distinguish when their intensity values are spread over a larger range.
- Subtle differences between neighbouring regions can become more visible, supporting inspection by a trained radiologist.
- A linear mapping is simple, computationally efficient, and easy to reproduce.
- The operation changes pixel values rather than discovering anatomy; interpretation must remain based on the original image and clinical context.

1.6 Alternative: gamma correction

Gamma correction is useful when a nonlinear adjustment is needed. For normalized intensity r in the range 0–1, the transformation is s = r^γ. If γ is less than 1, darker values are lifted more strongly; if γ is greater than 1, the image is darkened. For a dark image, γ = 0.7 may be tested after normalizing pixel values. The actual gamma should be selected carefully to avoid making noise or irrelevant structures appear important.

1.7 Limitations and validation

Excessive brightness or contrast can saturate highlights, hide subtle structures, and amplify noise. A single global linear transformation may not correct an image in which different regions have different illumination. Adaptive histogram equalization with contrast limiting (CLAHE) may be considered for local contrast, but it must be validated to avoid amplifying noise. Medical image enhancement should not replace the original image, alter diagnostic meaning, or be used without appropriate clinical review.

1.8 Algorithm

- Read the grayscale X-ray and confirm that pixel values are in the range 0–255.
- Choose a contrast gain a and brightness offset b based on the image histogram.
- For each pixel r, calculate s = clip[a(r − r_min) + b, 0, 255].
- Round the result to a valid integer intensity and store it in the output image.
- Compare original and enhanced images and check for clipping, amplified noise, and loss of detail.

1.9 Expected result

The output X-ray should be brighter and show improved separation of relevant grayscale structures. The numerical example demonstrates that input values 40, 60, 80, and 100 become 47, 74, 101, and 128. The improvement is useful only if important structures are preserved and the result remains suitable for clinical interpretation.

Q2 – HISTOGRAM PROCESSING AND SPATIAL FILTERING

2.1 Problem identification

A surveillance camera operating at night records images with poor contrast and unwanted noise. Dark foreground and background regions may have similar intensities, while random noise can create isolated bright or dark pixels. Two complementary operations are selected: histogram equalization for global contrast enhancement and median filtering for noise reduction.

2.2 Histogram equalization

A grayscale histogram counts how many pixels occur at each intensity. When the histogram is concentrated in a narrow range, the image may look flat or low-contrast. Histogram equalization uses the cumulative distribution of intensity values to remap the available values across a wider range.

For an image with L intensity levels, a common discrete mapping is:

$$s_k = \text{round}[(L - 1) \times \text{CDF}(r_k)]$$

The cumulative distribution function (CDF) is the cumulative probability of pixels with intensity less than or equal to r_k. For an 8-bit image, L = 256. Implementations may use a minimum-CDF adjustment so that the lowest occupied intensity maps to zero.

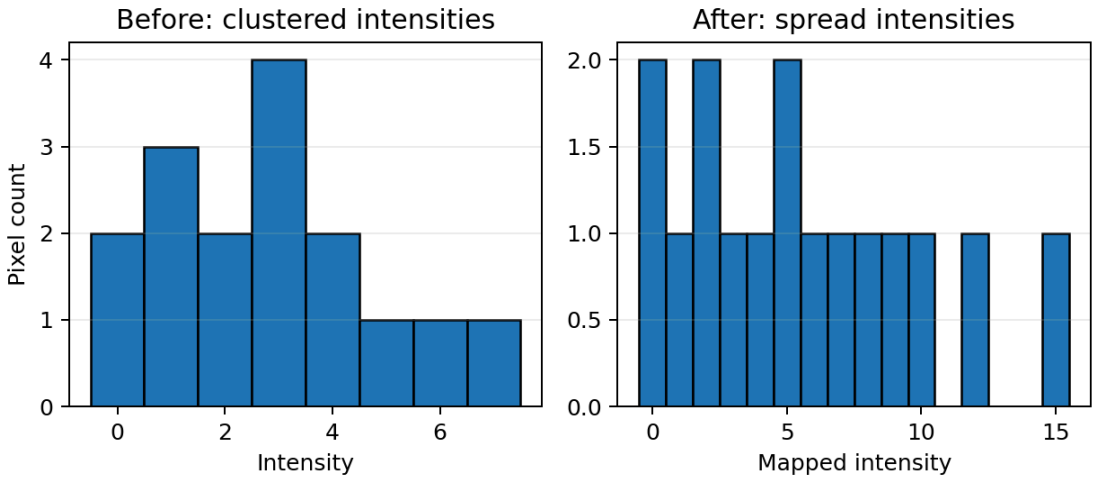


Figure 2. Conceptual histograms show a narrow distribution becoming more spread after remapping.

2.3 Illustrative histogram calculation

Consider a small 8-bit image whose normalized cumulative probability at an occupied grey level is 0.40. Under the simplified mapping, s = round(255 × 0.40) = round(102) = 102. If another grey level has CDF 0.75, its mapped value is round(255 × 0.75) = 191. These values illustrate the formula; a

real mapping is calculated from the actual image histogram, and discrete output values need not be evenly populated.

Original grey level	Illustrative CDF	Mapping	Output level
r_1	0.10	round(255 × 0.10)	26
r_2	0.40	round(255 × 0.40)	102
r_3	0.75	round(255 × 0.75)	191
r_4	0.95	round(255 × 0.95)	242

2.4 Selected spatial filter: median filter

A median filter is suitable for removing impulse-like noise, such as isolated black or white specks, while often preserving edges better than a simple averaging filter. For each pixel, the values in a local window (commonly 3 × 3) are sorted, and the middle value replaces the centre pixel.

Example 3 × 3 neighbourhood:

12	13	14
12	250	15
11	14	13

The nine values are 12, 13, 14, 12, 250, 15, 11, 14, and 13. Sorted in ascending order, they are 11, 12, 12, 13, 13, 14, 14, 15, 250. The median is 13. Therefore, the noisy centre value 250 is replaced by 13. This suppresses the isolated bright impulse while retaining a value consistent with its neighbours.



Figure 3. Median filtering replaces each pixel with the median of a local neighbourhood.

2.5 Recommended processing sequence

- Read the nighttime image and, if necessary, convert it to grayscale.
- Apply a median filter to reduce isolated impulse noise before contrast enhancement.
- Apply histogram equalization to spread intensity values and improve global contrast.
- Inspect the enhanced image and compare it with the original; adjust the filter window if details are lost.
- For strong non-uniform illumination, consider CLAHE or background correction instead of relying only on global equalization.

2.6 Comparison of the techniques

Aspect	Histogram equalization	Median filtering
Main purpose	Improve contrast by remapping intensity values.	Reduce local noise using neighbourhood statistics.
Operation type	Intensity mapping based on the whole-image histogram.	Spatial operation over a local window.
Best suited for	Images whose intensities occupy a narrow range.	Impulse noise and isolated outlier pixels.
Effect on edges	May make existing contrast boundaries more visible but does not directly denoise.	Often preserves step edges better than mean filtering.
Main risk	May amplify noise or over-enhance unimportant contrast.	Large windows can remove small objects and thin details.
Output	A contrast-enhanced grayscale image.	A spatially smoothed image with fewer outliers.

2.7 Why the combination is useful

The methods solve different problems. Median filtering reduces isolated noise, whereas histogram equalization improves the distribution of intensities. If equalization is applied to a very noisy image first, the noise may become more visible. Denoising first can therefore provide a cleaner input for contrast enhancement. However, the order and parameters should be tested because filtering can also remove small targets such as distant people or thin object boundaries.

2.8 Evaluation measures

- Visual inspection: determine whether people and objects are easier to distinguish from the background.
- Contrast measures: compare intensity spread or local contrast before and after processing.

- Noise measures: where a reference image is available, calculate mean squared error or peak signal-to-noise ratio (PSNR).
- Task performance: evaluate detection accuracy, missed detections, and false alarms on labelled surveillance frames.

2.9 Expected result

The median filter should reduce isolated noisy pixels, while histogram equalization should improve contrast between objects and their surroundings. The combined result is expected to be more useful for surveillance review, but low-light enhancement cannot recover details that were never captured by the camera.

Q3 – FREQUENCY-DOMAIN SMOOTHING AND SHARPENING

3.1 Problem identification

Satellite images contain useful geographical structures such as field boundaries, roads, shorelines, and changes in land cover. Noise often appears as rapid intensity fluctuations, while blurred boundaries reduce the sharpness of transitions. Frequency-domain filtering separates image information according to spatial frequency so that selected frequency components can be attenuated or emphasized.

3.2 Fourier transform concept

Let f(x, y) represent an image and F(u, v) its two-dimensional discrete Fourier transform (DFT). The transform expresses the image as a combination of spatial-frequency components. Low spatial frequencies generally correspond to slowly changing intensity patterns; high spatial frequencies generally correspond to fine details, sharp transitions, and also some forms of noise.

A frequency-domain filter forms G(u, v) = H(u, v)F(u, v), where H(u, v) is the filter transfer function. The inverse DFT then converts G(u, v) back into a spatial image g(x, y).

3.3 Low-pass filtering for smoothing

A low-pass filter allows lower spatial frequencies to pass and attenuates higher frequencies. It can reduce rapid noise fluctuations and smooth the image. A Gaussian low-pass filter is often preferred because its frequency response changes smoothly and tends to produce fewer ringing artefacts than an abrupt ideal cutoff.

A common Gaussian response is:

$$H_LP(u,v) = \exp[-D^2(u,v) / (2D_0^2)]$$

D(u, v) is the distance from the zero-frequency centre in the shifted Fourier spectrum, and D_0 controls the cutoff scale. A smaller D_0 gives stronger smoothing; a larger D_0 preserves more detail.

3.4 High-pass filtering for sharpening

A high-pass filter attenuates low spatial frequencies and emphasizes high-frequency components, making transitions and edges more prominent. One simple complementary definition is H_HP(u, v) = 1 − H_LP(u, v). A high-pass result may emphasize boundaries but can also amplify noise. A practical alternative is unsharp masking: blur the image, subtract the blurred version from the original to obtain a detail mask, then add a scaled version of that mask back to the original.

Unsharp masking can be expressed as $g(x,y) = f(x,y) + k[f(x,y) - f_{blur}(x,y)]$, where k controls sharpening strength. This method often preserves the overall image appearance while emphasizing edges.

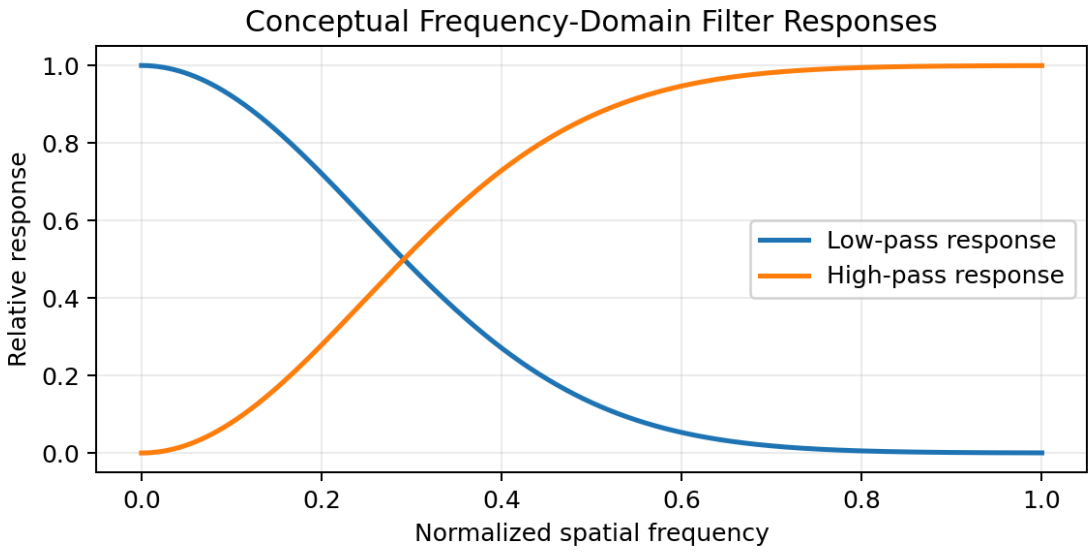


Figure 4. Conceptual responses: low-pass filtering retains low frequencies; high-pass filtering emphasizes higher frequencies.

3.5 Worked frequency-domain example

Assume a normalized frequency component has amplitude $F = 100$. At this frequency, suppose a selected low-pass filter has response $H_{LP} = 0.20$. Its output component is $G_{LP} = H_{LP} \times F = 0.20 \times 100 = 20$. The component is attenuated by 80%. For the complementary high-pass response $H_{HP} = 1 - 0.20 = 0.80$, the output component is $G_{HP} = 0.80 \times 100 = 80$. These calculations illustrate filter multiplication for one frequency component; the complete image contains many components and must be reconstructed with the inverse transform.

Filter	Example response H	Input component F	Output component G = HF	Interpretation
Low-pass	0.20	100	20	High-frequency component is strongly reduced.
High-pass	0.80	100	80	The same component is emphasized relative to low frequencies.

3.6 Processing algorithm

- Convert the satellite image to grayscale or process each colour channel appropriately.
- Compute the two-dimensional DFT and shift the zero-frequency component to the centre for convenient filter design.
- Construct a Gaussian low-pass filter H_LP using a selected cutoff D_0.
- Multiply the spectrum by H_LP, then apply the inverse DFT to obtain a smoothed image.
- For edge enhancement, construct a high-pass filter or use unsharp masking on the original image.
- Normalize the display range carefully and compare the outputs against the original image.

3.7 Differences and contribution to interpretation

Property	Low-pass filter	High-pass filter
Frequency effect	Retains slow intensity variation and suppresses high frequencies.	Suppresses slow variation and emphasizes rapid intensity changes.
Visual result	Smoother image with less fine noise.	Sharper-looking transitions and more visible boundaries.
Useful for	Reducing grain, small fluctuations, and some high-frequency noise.	Highlighting field edges, coastlines, roads, and boundaries.
Risk	Blurs thin roads and small geographical features.	Amplifies noise, creates halos, or exaggerates irrelevant edges.
Interpretation	Helps reveal broad land patterns when noise is distracting.	Helps locate boundaries but should be checked against original imagery.

3.8 Practical recommendation

Use low-pass filtering when noise prevents interpretation of broad land patterns. Use mild high-pass filtering or unsharp masking when boundaries are blurred. Applying strong smoothing and strong sharpening indiscriminately is not recommended: smoothing can erase the very boundaries later needed for segmentation, while sharpening can make noise appear to be a real geographical feature. A sensible workflow is to denoise conservatively, sharpen only as required, and retain the original image for comparison.

3.9 Expected result

The low-pass output should contain fewer high-frequency fluctuations and appear smoother. The high-pass or sharpened output should show stronger transitions at geographical boundaries. Together, the results support interpretation of both broad regions and local boundaries, provided the operator checks that the enhanced details are genuine.

Q4 – EDGE DETECTION AND THRESHOLD-BASED DEFECT SEGMENTATION

4.1 Problem identification

The manufacturing system must detect cracks and surface defects on metal components. Brightness varies across the image, so a single raw intensity value may not clearly separate defects from the background. The proposed approach combines noise reduction, Canny edge detection, thresholding, and basic morphological cleanup to generate a binary defect mask.

4.2 Selected edge detector: Canny

Canny edge detection is selected because it uses smoothing, gradient calculation, non-maximum suppression, and two thresholds with hysteresis. These steps help produce thin and connected edges while suppressing some noise. Before applying Canny, the image is normally converted to grayscale and lightly smoothed with a Gaussian filter.

4.3 Canny algorithm

- Smooth the grayscale image with a Gaussian kernel to reduce noise.
- Calculate intensity gradients in horizontal and vertical directions.
- Estimate gradient magnitude and direction.
- Apply non-maximum suppression to thin broad gradient responses into candidate edges.
- Use high and low thresholds to classify strong and weak edge pixels.
- Keep weak edges connected to strong edges through hysteresis and discard isolated weak responses.

Canny identifies intensity boundaries, not defects by itself. A scratch, a normal component edge, a reflection, and a crack may all generate edges. Additional location, shape, intensity, and size rules are required to classify which boundaries correspond to defects.

4.4 Thresholding and binary image construction

Thresholding assigns each pixel to foreground or background according to a threshold T. For a grayscale intensity I(x,y), a simple binary mask is:

$$B(x,y) = 1 \text{ if } I(x,y) < T; \text{ otherwise } B(x,y) = 0$$

This polarity is suitable only when the defect is darker than the background. If the defect is brighter, use $I(x,y) > T$. If brightness varies significantly across the image, adaptive thresholding or local background subtraction may be more reliable than a single global threshold.

4.5 Worked thresholding calculation

Assume a small region has grayscale values shown below and a dark-defect threshold $T = 80$. Pixels below 80 are labelled as candidate defects (1), while values of 80 or more are labelled background (0). The threshold is illustrative, not a universal value for metal inspection.

Original intensity image			
45	60	150	180
70	55	130	170
160	145	190	200
175	155	210	220

Binary mask (T = 80)			
1	1	0	0
1	1	0	0
0	0	0	0
0	0	0	0

The upper-left four pixels are below 80, so they are marked as foreground candidates. The remaining pixels are 80 or above and become background. In a practical system, the mask must be checked against the component geometry because dark stains, shadows, and texture may also fall below the threshold.

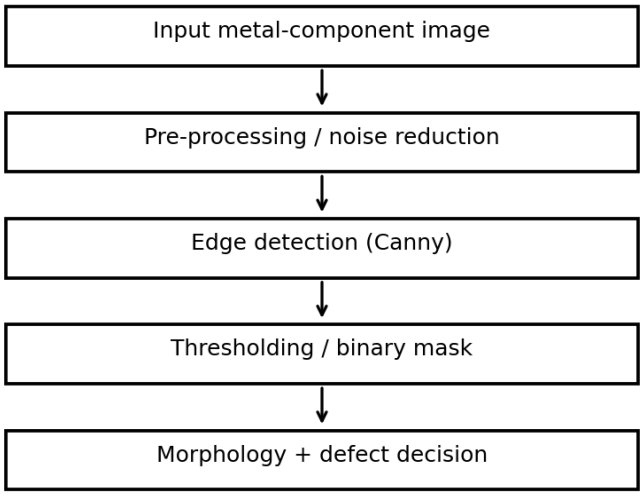


Figure 5. A practical defect-segmentation pipeline combines edge information and threshold-based region selection.

4.6 Combining edges and thresholding

Edge detection and thresholding provide complementary information. Canny highlights boundaries where intensity changes; thresholding selects pixels based on intensity. One practical workflow is to threshold a preprocessed image to create candidate dark regions, then use edges to refine their boundaries and separate adjacent candidates. Alternatively, edge maps can be used to close candidate contours before filling enclosed regions. The precise order depends on whether defects appear as dark regions, thin lines, or raised bright marks.

4.7 Morphological post-processing

- Opening removes small isolated foreground specks and can eliminate tiny false positives.
- Closing connects small gaps and may join broken parts of a crack boundary.
- Connected-component analysis labels each separate foreground region.
- Area, length, width, elongation, and boundary shape can help distinguish crack-like objects from stains.

Morphological operations must be chosen cautiously. An aggressive opening can remove a real thin crack, while excessive closing can merge nearby defects. A detected region should be compared with known defect sizes and the component's expected surface appearance.

4.8 Decision logic for automated inspection

Stage	Operation	Purpose
1	Image acquisition and calibration	Maintain consistent lighting, focus, and scale.
2	Denoising and illumination correction	Reduce noise and brightness variation.
3	Canny edge detection	Locate candidate boundaries and crack-like transitions.
4	Thresholding	Create a binary mask of candidate defect pixels.
5	Morphology and connected components	Repair small gaps and isolate candidate regions.
6	Feature measurement and decision	Compare size, shape, position, and quality limits.
7	Human review / traceable output	Verify uncertain detections and record the inspection result.

4.9 Evaluation

Use labelled samples of defective and non-defective components to measure precision, recall, and false-positive rate. Precision indicates how many detected defects are genuine; recall indicates how many real defects are found. False negatives are particularly important in safety-critical manufacturing because a missed crack may allow a faulty component to pass inspection. Thresholds and Canny parameters should be selected using representative images, not only one example.

4.10 Expected result

The output binary mask should highlight candidate defect regions while suppressing most background pixels. Canny contributes boundary information, thresholding separates candidate pixels, and morphology improves region continuity. The final decision should use validated defect criteria and retain an option for human review.

Q5 – REGION-BASED SEGMENTATION AND BOUNDARY DETECTION

5.1 Problem identification

An agricultural monitoring system must separate crop fields, water bodies, and surrounding land in aerial imagery. Some neighbouring regions have similar intensity values, and boundaries may be discontinuous because of shadows, vegetation, sensor noise, or low contrast. Region-based segmentation groups pixels with similar characteristics, while edge linking and boundary detection attempt to create coherent borders between the resulting regions.

5.2 Region-based segmentation

Region-based methods assign neighbouring pixels to a region when they satisfy a similarity rule. Similarity can be based on grayscale intensity, colour, texture, vegetation index, or a combination of features. Two common approaches are region growing and split-and-merge segmentation.

5.3 Selected method: seeded region growing

Seeded region growing starts with one or more seed pixels chosen inside known examples of a crop field, water body, or land region. Neighbouring pixels are added when their feature values are sufficiently similar to the current region model. A simple grayscale rule is $|I(x,y) - \mu_R| \leq \tau$, where μ_R is the mean intensity of region R and τ is a selected tolerance.

5.4 Worked region-growing example

Suppose a seed lies in a crop region with mean grayscale intensity $\mu_R = 120$, and the similarity tolerance is $\tau = 15$. A neighbouring pixel with intensity 128 has $|128 - 120| = 8$, which is less than or equal to 15, so it can be added to the region if it is connected to the growing boundary. A neighbouring pixel with intensity 150 has $|150 - 120| = 30$, so it is not added under this rule. A pixel with intensity 105 differs by 15 and is accepted at the tolerance boundary.

Candidate pixel intensity	Absolute difference from 120	Rule $\|I - \mu R\| \leq 15$	Decision
128	8	$8 \leq 15$	Accept
150	30	$30 > 15$	Reject
105	15	$15 \leq 15$	Accept
118	2	$2 \leq 15$	Accept

5.5 Region growing procedure

- Preprocess the aerial image to reduce noise and correct severe illumination differences.
- Select reliable seed points or initial regions for the target land-cover classes.
- Calculate the current region mean or feature model.
- Examine neighbouring pixels and add those satisfying the similarity threshold.
- Update the region model and continue until no eligible neighbours remain.
- Repeat for other classes and resolve overlaps or unlabelled pixels.
- Inspect the region map and refine parameters using ground-truth samples where available.

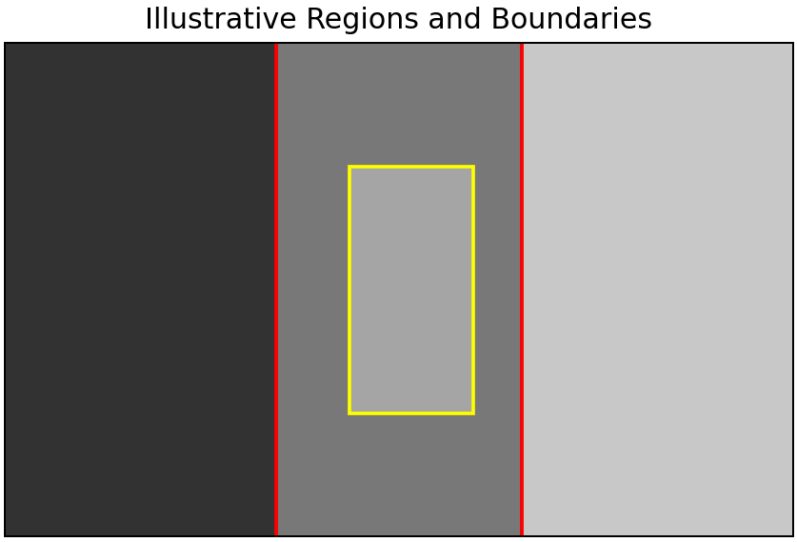


Figure 6. Illustrative image containing three broad regions and a smaller enclosed region; boundaries are overlaid for clarity.

5.6 Edge linking for discontinuous boundaries

Edge detectors may produce short broken segments rather than a complete border. Edge linking attempts to connect segments that are likely to belong to the same boundary. Candidate segments can be compared using spatial proximity, gradient direction, orientation, and intensity continuity. Endpoints that are close together and have compatible directions are more likely to be linked.

- Local linking connects nearby edge pixels when their direction and magnitude are compatible.
- Morphological closing can connect small gaps in a binary edge map.
- Contour tracing follows connected boundary pixels to create an ordered outline.
- More advanced methods can use graph-based optimization or active contours when boundaries are weak or curved.

5.7 Boundary detection and region separation

After region growing, a boundary can be identified wherever adjacent pixels have different region labels. If L(x,y) is a region-label image, a simple boundary condition is that at least one neighbour has a label different from L(x,y). The resulting boundary map can be refined by removing tiny isolated fragments and tracing closed contours.

A complete boundary helps estimate field area, perimeter, shape, and adjacency to water or surrounding land. However, a boundary should not be forced closed if the image does not support that conclusion; incorrect closure can merge different land regions or invent a field outline.

5.8 Combining region similarity and edge evidence

Region similarity is useful for filling a whole field or water body, while edges are useful for locating the transition between adjacent regions. Combining them reduces dependence on either source alone. For example, region growing can identify a coherent crop area, and a gradient-based boundary map can refine the edge where the crop meets a water channel. If an edge is weak, the region labels and local feature statistics can guide the boundary; if two regions have similar intensity, texture or multispectral information may be needed.

5.9 Challenges and controls

Challenge	Possible effect	Control / mitigation
Similar intensities in different classes	Unwanted merging of crop and bare land.	Add colour, texture, vegetation indices, or contextual features.
Shadows and uneven lighting	A single similarity threshold may split one field.	Apply illumination correction or use adaptive feature models.
Broken edges	Boundary map contains gaps.	Use cautious edge linking, closing, and contour validation.
Incorrect seed points	Region grows into the wrong land cover.	Use multiple validated seeds or automatic seed selection.
Noise and small objects	Over-segmentation into many tiny regions.	Denoise gently and remove components below justified size limits.
Mixed pixels at field borders	Uncertain region membership.	Use higher-resolution imagery or soft/probabilistic classification.

5.10 Evaluation and expected result

Compare the generated region map with manually labelled or verified land-cover data. Useful measures include intersection over union (IoU), Dice coefficient, boundary precision and recall, and area error.